

ABSTRACT

Virtual Memory Manager Simulator that demonstrates how different page replacement algorithms such as FIFO, LRU, and Optimal Replacement handle memory management tasks. Users can input custom page reference strings and memory frame counts to observe detailed, real-time simulation outputs including page hits, page faults, and memory states. The simulator's user-friendly web and desktop interfaces, combined with visual outputs and statistical metrics like hit rate and page fault count, simplify complex memory management concepts. Designed as a practical learning tool, it bridges the gap between theoretical understanding and real-world memory management techniques.